

KARTHIK B

Robotics Software Engineer

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PROFILE SUMMARY

Robotics Software Engineer with 3 years of experience building and deploying software for industrial automation and warehouse robotics systems. Hands-on experience in Python, ROS2, PLC integration, Modbus communication, and backend systems supporting ASRS, AMR, SCARA, and conveyor-based automation.

SKILLS

Robotics & Automation Software: Robotics Testing, Robotics Automation, ROS2, ASRS, AMR, SCARA Robots, Conveyor Systems, Nano-Warehouse Automation

Control Systems & Communication: PLC Integration, Modbus (RS485), PyPLC Library, Arduino Programming, Sensor Calibration, Industrial Cameras and routers

Programming & Backend: Python, FastAPI, REST API Development, MQTT, SQL, PostgreSQL, Backend Integration

Deployment & Infrastructure: Linux, Docker, On-Premise Deployments, Server Deployment, SSH, tmux, Industrial Routers

Monitoring & Reliability: System Monitoring, Logging, Prometheus, Grafana, Production Troubleshooting

Networking (Industrial / Basic): IP Addressing, Router Configuration (Basic), Port Configuration, Network Troubleshooting

Supporting Frontend: HTML5, CSS3, JavaScript (Basic), Bootstrap, Frontend UI Support

WORK EXPERIENCE

Software Engineer – Robotics & Automation Systems – Leapmile Robotics

August 2023 – Present

- Developed and maintained Python-based software systems supporting ASRS, AMR, SCARA, and conveyor-based nano-warehouse automation
- Built backend services and REST APIs using Python and FastAPI to support robot control, monitoring, and operational dashboards
- Led robotics testing and automation workflows, including system validation, sensor calibration, and production readiness checks
- Developed a custom PyPLC library to enable reliable PLC communication over Modbus (RS485). Integrated PLCs, Arduino controllers, sensors, and industrial cameras with robotics software systems
- Implemented MQTT-based asynchronous communication for real-time data exchange between robots, controllers, and backend services. Deployed robotics software on on-premise Linux production systems using Docker, SSH, and tmux
- Configured and supported industrial routers, robot PCs, and networked devices used in warehouse automation environments. Monitored and troubleshot robotics software and backend systems using logs, metrics, and Linux CLI tools
- Collaborated with cross-functional teams across robotics hardware, backend, and operations to ensure stable production deployments

Web Development Intern – BasketHunt Pvt Ltd

January 2023 – July 2023

- Learned and implemented basic web page designs using HTML, CSS, and JavaScript, and supported simple frontend development tasks under senior developer guidance

GPS Operations Partner – BlackBuck Logistics

2021 – 2022

- Supported GPS-based fleet operations and basic system coordination

EDUCATION

Bachelor of Engineering – Automobile Engineering

June 2016 – November 2020

Paavai College of Engineering, Anna University | Percentage: 70%

PROJECTS

- **ROS 2 Publish–Subscribe Demo:** Built and tested ROS 2 Python and C++ publisher–subscriber nodes to demonstrate inter-node communication using rclpy and rclcpp. Created ROS 2 workspaces and packages, built them using colcon, and validated node execution and messaging using ROS 2 CLI tools and rqt_graph. https://github.com/karthik0497/ros2_developments
- **Crypto PriceLoop – Telegram Bot:** Built an asynchronous Python-based Telegram bot for real-time crypto price alerts, portfolio tracking, and market updates using CoinMarketCap APIs, PostgreSQL, and Docker. https://github.com/karthik0497/telegram_crypto_notification_bot | https://t.me/Crypto_priceloop_bot

CERTIFICATIONS

- **Postman API Fundamentals – Student Expert** – Postman (REST APIs, HTTP methods, request/response handling, authentication)
- **Python Web Development** – Live Wire Institute (Python, Django, HTML, CSS, JavaScript, Bootstrap)

LANGUAGES

Tamil (Native), English (Professional), Malayalam (Beginner)